

FRYEBURG EASTERN SLOPE, ME, USA

WMO: 726183

Lat: 43.991N

Lon: 70.948W

Elev: 445

StdP: 14.46

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 54772

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-3.7	-19.5	1.9	-6.8	-14.1	2.6	0.4	19.4	22.8	17.6	22.9	1.2	280	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	89.7	71.7	86.1	69.8	82.9	68.2	74.4	84.1	72.7	82.0	70.9	79.3	6.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	119.4	78.4	70.0	112.3	77.0	68.1	105.0	75.4	38.1	84.3	36.6	81.6	35.0	79.0	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.4	13.5	11.7	-20.7	94.3	6.6	2.4	-25.4	96.1	-29.3	97.5	-33.0	98.9	-37.8	100.7		
			-21.3	77.7	6.5	2.2	-26.0	79.3	-29.8	80.5	-33.5	81.8	-38.2	83.3		
			DB													
			WB													

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	19.0	22.1	30.7	43.3	55.1	63.7	69.1	67.3	59.8	47.5	36.4	25.4	(6)
(7)		DBStd	19.37	11.88	10.32	10.15	8.18	7.55	6.85	5.59	5.57	7.55	7.95	8.85	10.15	(7)
(8)		HDD50	3942	960	782	603	235	31	1	0	0	8	144	414	764	(8)
(9)		HDD65	7649	1425	1202	1064	653	320	106	23	38	190	542	857	1229	(9)
(10)		CDD50	2145	0	0	5	33	189	411	591	538	303	68	7	0	(10)
(11)		CDD65	377	0	0	0	1	14	66	149	111	35	1	0	0	(11)
(12)		CDH74	4513	0	0	10	53	332	837	1617	1229	406	29	0	0	(12)
(13)		CDH80	1401	0	0	2	13	103	264	529	376	112	2	0	0	(13)
(14)	Wind	WSAvg	3.8	3.8	4.2	4.8	4.9	4.5	3.8	3.2	3.0	3.0	3.6	3.7	3.6	(14)
(15)	Precipitation	PrecAvg	48.0	3.2	3.4	4.1	3.9	3.7	4.6	4.3	3.9	4.2	4.4	4.0	4.4	(15)
(16)		PrecMax	65.1	7.0	8.7	9.5	10.1	9.3	9.3	7.3	7.6	10.9	15.0	5.8	7.2	(16)
(17)		PrecMin	31.7	0.5	1.3	1.0	0.0	0.6	2.1	1.6	0.9	0.8	0.8	0.7	1.7	(17)
(18)		PrecStd	8.9	1.3	1.6	2.1	2.5	2.2	2.0	1.7	1.7	2.6	3.2	1.3	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	53.2	70.2	81.1	89.3	91.8	92.5	92.5	89.6	78.5	65.8	56.2	(19)
(20)		0.4%	MCWB	48.1	44.0	54.1	59.4	66.7	72.5	74.1	72.1	72.6	62.2	56.2	49.1	(20)
(21)		2%	DB	44.7	46.2	57.2	72.3	82.3	87.4	89.0	88.0	82.6	72.0	60.7	48.2	(21)
(22)		2%	MCWB	39.4	39.1	45.8	54.5	62.5	69.6	72.3	71.3	68.8	60.1	52.5	43.0	(22)
(23)		5%	DB	39.5	41.3	50.6	65.5	76.9	82.2	85.7	84.0	78.7	66.2	55.3	43.6	(23)
(24)		5%	MCWB	34.4	35.3	41.0	51.0	59.7	66.7	70.2	69.4	66.6	57.4	49.4	38.3	(24)
(25)		10%	DB	35.8	37.2	45.7	60.2	72.3	79.0	82.4	81.1	74.5	61.9	50.9	39.2	(25)
(26)		10%	MCWB	31.8	32.2	38.4	47.3	57.1	64.6	68.7	67.9	63.9	54.5	45.2	35.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	46.2	55.2	62.6	69.2	74.9	76.9	76.1	74.5	66.7	60.5	51.1	(27)
(28)		0.4%	MCDWB	51.7	50.5	66.1	73.5	82.7	86.3	87.9	85.4	85.1	72.0	64.0	54.8	(28)
(29)		2%	WB	40.2	40.0	48.0	58.0	65.8	72.4	74.7	73.8	71.3	62.3	54.7	44.4	(29)
(30)		2%	MCDWB	43.7	44.7	56.3	69.5	77.4	83.1	84.0	83.1	79.6	68.3	57.8	48.2	(30)
(31)		5%	WB	35.5	36.3	42.5	53.0	62.8	69.7	72.9	71.9	69.0	59.0	50.0	39.3	(31)
(32)		5%	MCDWB	39.1	40.6	48.8	61.2	72.4	78.5	81.8	80.2	75.2	65.0	55.1	42.5	(32)
(33)		10%	WB	32.2	33.2	38.8	48.6	59.7	67.3	71.1	70.1	66.3	55.7	46.1	35.9	(33)
(34)		10%	MCDWB	35.0	36.4	44.9	58.9	69.1	75.3	79.5	77.3	72.2	60.8	49.7	39.1	(34)
(35)	Mean Daily Temperature Range		MDBR	21.9	22.6	21.5	23.8	24.8	23.3	23.7	24.6	25.2	22.3	19.3	19.4	(35)
(36)		5% DB	MCDBR	21.4	24.1	28.7	36.7	35.0	29.6	27.4	28.8	28.9	28.5	25.9	22.7	(36)
(37)		5% DB	MCWBR	17.4	18.1	18.7	20.2	17.2	13.9	12.6	13.7	15.9	17.9	18.8	18.1	(37)
(38)		5% WB	MCDBR	19.7	21.8	25.5	30.8	28.6	25.4	23.5	23.2	23.5	24.0	22.6	20.5	(38)
(39)		5% WB	MCWBR	17.2	17.0	18.1	19.5	15.3	12.9	11.9	12.4	13.9	16.6	18.0	17.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.268	0.288	0.318	0.349	0.377	0.428	0.443	0.411	0.368	0.328	0.309	0.280	(40)	
(41)		taud	2.425	2.344	2.380	2.412	2.382	2.247	2.214	2.297	2.403	2.521	2.528	2.481	(41)	
(42)		Ebn at Noon	276	288	292	291	284	268	262	267	272	271	259	261	(42)	
(43)		Edh at Noon	23	30	33	35	37	43	44	39	32	25	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	508	785	1147	1424	1592	1723	1781	1621	1317	822	536	398	(44)	
(45)		RadStd	43	69	70	158	195	148	135	91	82	79	55	38	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page